DDA2001: Introduction to Data Science

Mid-term Exam
SDS, CUHK(SZ)

1:30pm - 3:30pm, March 16, 2024

Instructions

1. The time duration is **120 minutes**.
2. The total mark is **100 points**.
3. There are **7 questions** in total, please check.
4. Please write down your calculations and statements **clearly and in detail**.
5. Give answers up to **2 decimal places** where appropriate.
6. You should answer the questions in the answer sheet in **sequential order**.
7. Please submit the test paper and answer sheet **together** at the end.
8. The time for checking the answer sheets is **6pm-8pm on March 22nd at TXB 604**. Once you miss it, it is not allowed to check it again for any reason.
9. You are allowed to bring any dictionary without formulas and a non-programmable calculator to the exam. Bringing any other materials or electronic devices will be considered cheating. Students caught cheating on exams or when checking the answer sheets, will ultimately fail the course and are not approved to late drop.

Declare: I have read the instructions carefully. I will fully comply with the exam rules and instructions.

Name: _____ Student ID: _____

- The probability density function of the normal distribution with mean μ and variance σ^2 is as follows:

$$f(x|\mu,\sigma) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}.$$

- If $Y \sim \mathcal{N}(\beta_0 + \beta_1 X, \sigma^2)$, the MLE of β_0 and β_1 is as follows:

$$\hat{\beta}_1 = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sum_i (x_i - \bar{x})^2} \text{ and } \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}.$$

1. **Single Choice Questions:**

   (a) **(2 points)** Sometimes we do not have access to the entire data set (population) and we have to infer our conclusions using sample data. Which of the following approaches addresses working with sample data to conclude about the population?
   A. Probability theory
   B. Statistics
   C. Simulation
   D. Optimization

   Solution: B

   (b) **(2 points)** What is the primary goal of Maximum Likelihood Estimation (MLE)?
   A. Minimize the variance of a probability distribution
   B. Estimate the mean of a given distribution
   C. Find the parameter values that maximize the likelihood of observed data
   D. Compute the exact probability of a single event

   Solution: C

   (c) **(2 points)** Which of the following best describes the probability density function (PDF) for a continuous random variable?
   A. It gives the probability of a specific value occurring, $P(X = x)$
   B. The probability of a specific value is always zero, and the PDF describes how likely the variable is to fall within a small interval
   C. It represents the cumulative probability up to a certain value
   D. It is a function used to approximate discrete distributions

   Solution: B

   (d) **(2 points)** If events A and B are mutually exclusive, which of the following statements is true?
   A. $P(A \cap B) = P(A) \times P(B)$
   B. $P(A \cup B) = P(A) + P(B)$
   C. $P(A \mid B) = P(B \mid A)$
   D. $P(A \cap B)$ is greater than zero

   Solution: B

(e) **(2 points)** If A and B are independent events, which of the following must be true?
A. $P(A \mid B) = 1$
B. $P(A) = P(B)$
C. $P(A \cap B) = P(A) + P(B)$
D. $P(A \cap B) = P(A)P(B)$

Solution: D

(f) **(2 points)** Which of the following describes conditional probability?
A. The probability of an event occurring given that another event has already occurred
B. The probability of two independent events occurring simultaneously
C. The probability of a mutually exclusive event
D. The total probability of all possible outcomes

Solution: A

(g) **(2 points)** The central limit theorem states that:
A. Any sample will always follow a normal distribution
B. The sampling distribution of the sample mean approaches normality as the sample size increases
C. The mean of a sample is always equal to the mean of the population
D. The probability of extreme events increases with larger sample sizes

Solution: B

(h) **(2 points)** You have a coin that you think is biased. You flip it 4 times and get the sequence HHHT. What is the maximum likelihood estimate for the probability of getting heads?
A. 1/2
B. 2/3
C. 3/4
D. 4/5

Solution: C

(i) **(2 points)** What does σ^2 represent in the linear regression model $Y \sim N(\beta_0 + \beta_1 X, \sigma^2)$?
A. The mean of the dependent variable Y.
B. The slope of the regression line.
C. The variance of the independent variable X.
D. The variance of the residual error term in the regression model.

Solution: D

(j) **(2 points)** In a normal distribution, what happens when the standard deviation increases?
A. The peak of the distribution increases.

B. The distribution becomes steeper.
C. The distribution becomes flatter.
D. The shape of the distribution remains unchanged.

Solution: C

2. **(15 points)** Assume the days of the year can be numbered 1 to 365. Assume that birthdays are equally likely to fall on any day of the year. Consider a group of n people, of which you are not a member. An element of the sample space S will be a sequence of n birthdays (one for each person).

   (a) (**3 points**) Define the probability function P for S. (This will depend on n.)

   (b) (**6 points**) Consider the following events:
   (A) "someone in the group shares your birthday"
   (B) "some two people in the group share a birthday"
   (C) "some three people in the group share a birthday"
   Carefully describe the subset of S that corresponds to each event.

   (c) **(3 points)** Find an exact formula for $P(A)$.

   (d) **(3 points)** Let n^* be the smallest n such that $P(A) > 0.5$. Justify why n^* is greater than $365/2$ without doing any computation.

Solution:

(a) The sample space S is the set of all sequences of n birthdays. That is, all sequences

$$\omega = (b_1, b_2, b_3, \ldots, b_n),$$

where each entry is a number between 1 and 365. There are 365^n sequences of n birthdays. Since they are all equally likely, $P(\omega) = \frac{1}{365^n}$ for every sequence ω. (3 points)

(b) Event A: Suppose my birthday is on day b. Then "an outcome ω is in A" is equivalent to "b is in the sequence for ω", i.e. $b = b_k$ for some index k between 1 and n. More symbolically, an outcome ω is in A if and only if $b_k = b$ for some index $k \in \{1, \ldots, n\}$. (2 point)

Event B: "An outcome ω is in B" is equivalent to "two of the entries in ω are the same". That is, an outcome ω is in B if and only if $b_j = b_k$ for two (different) indices $j, k \in \{1, \ldots, n\}$. (2 point)

Event C: an outcome ω is in C if and only if $b_j = b_k = b_l$ for three (distinct) indices $j, k, l \in \{1, \ldots, n\}$. (2 point)

(c) It's easier to calculate $P(A^c)$. Since there are 364 birthdays that are not mine, there are 364^n outcomes in A^c. Hence

$$P(A) = 1 - P(A^c) = 1 - \frac{364^n}{365^n}.$$

(3 points)

(d) Ignoring the fractions, $\frac{365}{2}$ different birthdays would have a 50 percent chance of matching your birthday. But, $\frac{365}{2}$ people probably don't all have different birthdays, so they have a less than 50 percent chance of matching. (3 points)

$\square$

3. **(25 points)** The NBA teams are adopting a new technology for training. They would like to know how much improvement there is on the number of threes made during a game.

   (a) **(5 points)** Before the new technology was adopted, the numbers of threes made during NBA games were 14, 26, 25, and 13. Assume the number of threes made during an NBA game follows a Poisson distribution. Calculate the maximum likelihood estimate for the parameter λ. (Hint: The parameter λ shall be positive. If the parameter is λ, the probability mass function of a Poisson random variable $X = k$ is $\frac{\lambda^k e^{-\lambda}}{k!}$.)

   (b) **(5 points)** After the new technology was adopted, the numbers of threes made during NBA games were 20, 24, 28, and 15. What is the maximum likelihood estimate for the new parameter $\tilde{\lambda}$? Again assume the number of threes made follows Poisson distribution.

   (c) **(5 points)** Calculate the mean of a Poisson random variable with parameter λ. (Hint: A poisson random variable can take any integer value starting from 0. Also, you may use the power series expansion of the exponential function $e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}$.) Based on the previous two MLE estimates, what is the improvement on the mean number of threes after introducing the new training technology?

   (d) **(10 points)** If we use a linear regression model to estimate the improvement on the average number of threes after introducing the new training technology, what would be your answer? (Hint: Assume that $Y \sim \mathcal{N}(\beta_0 + \beta_1 X, \sigma^2)$, where $X = 1$ if the new technology is used and $X = 0$ if the new technology is not used, Y is the corresponding number of threes.)

Solution:

(a) The likelihood function is

$$L(\lambda) = \frac{\lambda^{78} e^{-4\lambda}}{14!26!25!13!}.$$

Therefore the log-likelihood function is

$$l(\lambda) = 78 \log \lambda - 4\lambda - \log(14!26!25!13!),$$

with derivative

$$l'(\lambda) = \frac{78}{\lambda} - 4.$$

Setting $l'(\lambda) = 0$, we get the MLE for λ is 19.5.

(b) The likelihood function is

$$L(\tilde{\lambda}) = \frac{\tilde{\lambda}^{87} e^{-4\tilde{\lambda}}}{20!24!28!15!}.$$

Therefore the log-likelihood function is

$$l(\tilde{\lambda}) = 87 \log \tilde{\lambda} - 4\tilde{\lambda} - \log(20!24!28!15!),$$

with derivative

$$l'(\tilde{\lambda}) = \frac{87}{\lambda} - 4\tilde{\lambda}.$$

Setting $l'(\tilde{\lambda}) = 0$, we get the MLE for $\tilde{\lambda}$ is 21.75.

(c) The mean of a Poisson random variable X with parameter λ is

$$\begin{aligned}
\mathbb{E}X &= \sum_{k=0}^{\infty} k \cdot \frac{\lambda^k e^{-\lambda}}{k!} \\
&= \sum_{k=1}^{\infty} k \cdot \frac{\lambda^k e^{-\lambda}}{k!} \\
&= \lambda e^{-\lambda} \sum_{k=1}^{\infty} \frac{\lambda^{k-1}}{(k-1)!} \\
&= \lambda e^{-\lambda} e^{\lambda} \\
&= \lambda.
\end{aligned}$$

The improvement on λ after introducing the new training technology is $21.75 - 19.5 = 2.25$.

(d) The data for linear regression is $((0,14),(0,26),(0,25),(0,13),(1,20),(1,24),(1,28),(1,15))$. Denote the ith data by (x_i, y_i). Then $\bar{x} = 0.5$ and $\bar{y} = 20.625$. The MLE of β_1 is

$$\begin{aligned}
\hat{\beta}_1 &= \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sum_i (x_i - \bar{x})^2} \\
&= \frac{4.5}{2} \\
&= 2.25.
\end{aligned}$$

Therefore the improvement on the average number of threes after introducing the new technology is 2.25.

□

4. **(10 points)** Demonstrate how random variables can be utilized to approximate the integral $\int_0^1 g(x)dx + \int_2^5 g(x)dx$, where $g(\cdot)$ is a known and bounded function. Be sure to specify the theorem that justifies the approximation.

   Solution: We can use the following steps to approximate the integral. For instance $\int_0^1 g(x)dx$, we do followings steps:

   - Generate uniformly distributed random numbers between 0 and 1. These will represent your x values. (3 point)
   - Evaluate the function $f(x)$ for each of these random x values. (3 point)
   - Take the average of all these function values and multiply it by the interval length, where we used the fact that expectation can be approximated by long-run average (or just Law of Large Numbers). (4 point)

□

5. (a) **(3 points)** What is the difference between a probability mass function (PMF) and a probability density function (PDF) in the context of random variables?

   (b) **(7 points)** Explain the difference between point and interval estimators, and provide three real-life examples where they are used.

   Solution:

   (a) In a PMF, each possible outcome of a discrete random variable is assigned a probability between 0 and 1, such that the sum of all probabilities equals 1. On the other hand, a PDF describes the likelihood of a random variable falling within a certain range of values.

   (b) A point estimation gives a single value estimate of a population parameter, and it provides no information about the variability of the estimate. On the other hand, an interval estimation gives a range of values within which the population parameter is expected to fall with a certain level of confidence.

□

6. The random variable X has the following cumulative distribution function:

$$F(x) = \begin{cases} 0 & x < -2, \\ \alpha & -2 \leq x < 0, \\ 8\alpha^2 & 0 \leq x < 1, \\ 1 & x \geq 1. \end{cases}$$

where α is unknown. Suppose $E[X] = 0$. Then,

(a) **(5 points)** What is the probability mass function for X.

(b) **(5 points)** Compute $E[X^2 + |X - 1|]$.

Solution:

(a) Since $E[X] = 0$, we have

$$-2\alpha + (1 - 8\alpha^2) = 0,$$

which implies that $\alpha = \frac{1}{4}$. Hence, $f(-2) = \frac{1}{4}$, $f(0) = \frac{1}{4}$, $f(1) = \frac{1}{2}$.

(b)

$$E[X^2 + |X - 1|] = 7 * \frac{1}{4} + 1 * \frac{1}{4} + 1 * \frac{1}{2} = \frac{5}{2}.$$

□

7. **(10 points)** A total of n people randomly take their seats around a circular table with n chairs. **No** two people have the same height. What is the expected number of people who are shorter than both of their immediate neighbors?

As an illustration, we show a figure below: each person has two immediate neighbors—one on their left and one on their right.

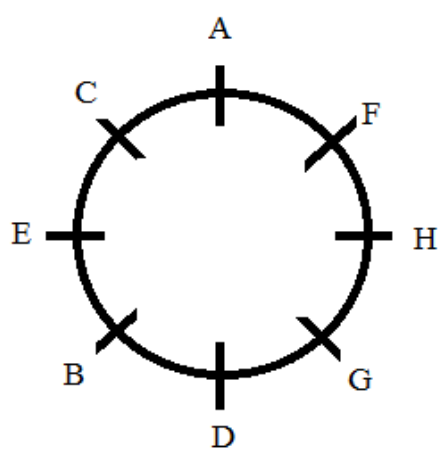

Figure 1: Illustration of immediate neighbors: For example, person A's immediate neighbors are C and F; person F's immediate neighbors are A and H, and so on.

Solution: Label the seats 1 to n going clockwise around the table. Let X_i be the Bernoulli random variable with value 1 if the person in seat i is shorter than their neighbors. Then

$$\sum_{i=1}^{n} X_i$$

represents the total number of people who are shorter than both of their neighbors, and, by linearity of expected value,

$$E[X] = E\left[\sum_{i=1}^{n} X_i\right] = \sum_{i=1}^{n} E[X_i].$$

Among 3 random people, the probability that the middle one is the shortest is $\frac{1}{3}$. Therefore, $X_i \sim \text{Bernoulli}\left(\frac{1}{3}\right)$, which implies $E[X_i] = \frac{1}{3}$. Therefore, the expected number of people shorter than both their neighbors is

$$E[X] = \sum_{i=1}^{n} E[X_i] = \frac{n}{3}.$$

(4 points for accurately identifying the random variables; 4 points for noting the linearity of expectation (even if the initial step is incorrect); 2 points for correctly calculating the expectation) $\square$